



Вселенная

Практическое применение машинного обучения в компьютерном зрении

Авторы: Валлиappa Лакшманан, Мартин Гёрнер, Райан Гиллард

Тип файла: pdf

Размер: 20,9 МБ

Язык: Английский

Страниц: 480

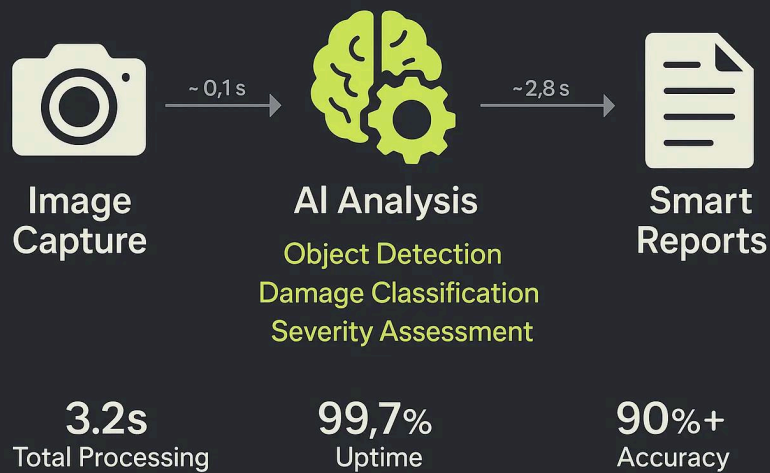
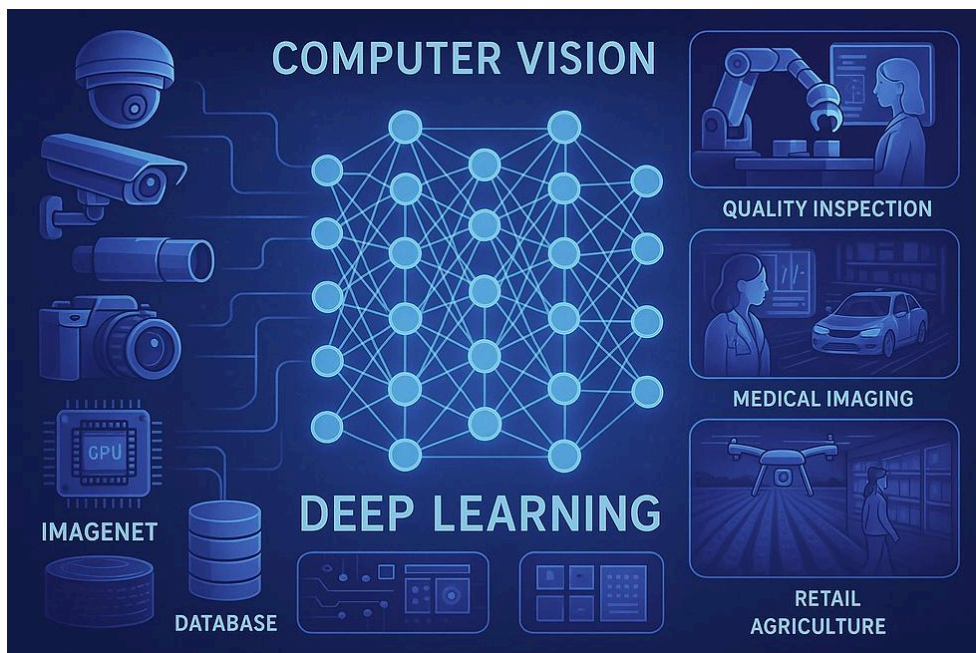
Практическое машинное обучение для компьютерного зрения: полное инженерное руководство по созданию интеллектуальных систем машинного зрения 🤖 👁 📷

Введение 🚀

Компьютерное зрение изменило то, как машины воспринимают окружающий мир. От беспилотных автомобилей 🚗 и медицинской визуализации 🏥 до распознавания лиц 😊 и промышленного контроля 🏭 — машинное обучение позволяет компьютерам с поразительной точностью интерпретировать изображения и видео.

Традиционная обработка изображений опиралась на разработанные вручную правила и математические алгоритмы. Современные **машинное обучение (Machine Learning, ML)** и **глубокое обучение (Deep Learning, DL)** позволяют компьютерам автоматически выявлять особенности на миллионах изображений, что делает системы машинного зрения более точными, масштабируемыми и адаптируемыми.

В этом руководстве практическое машинное обучение для компьютерного зрения рассматривается с инженерной точки зрения. Независимо от того, новичок ли вы в области классификации изображений или опытный инженер, создающий продукты на основе искусственного интеллекта, в этой статье вы найдете информацию о концепциях, рабочем процессе, сравнениях, примерах, трудностях и лучших практиках, необходимых для создания успешных приложений в области компьютерного зрения.



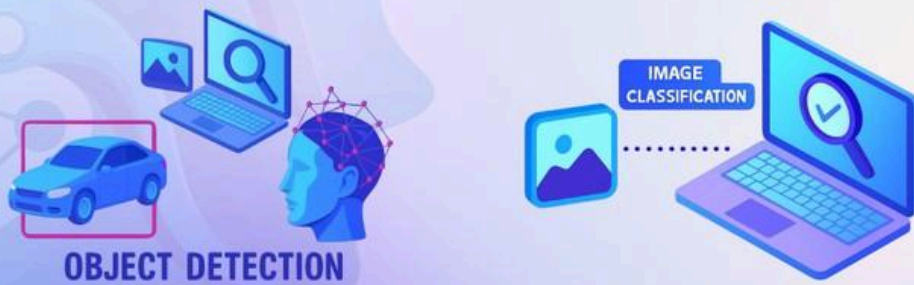
How AI Person Detection Technology Works

Smart human recognition & false alarm reduction

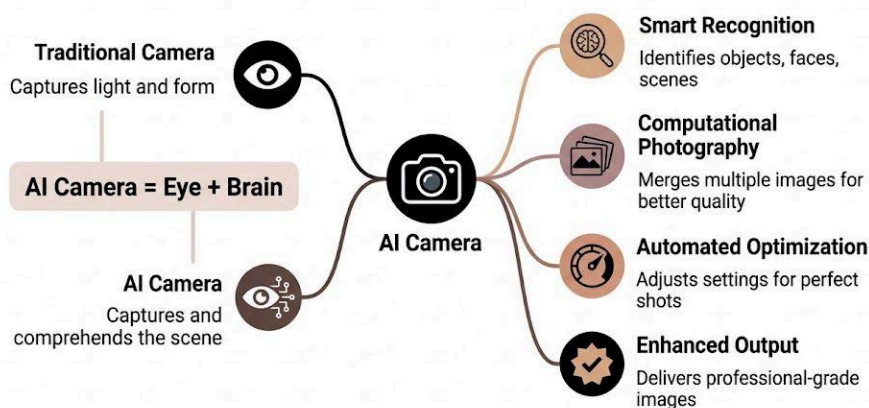


COMPUTER VISION

Building Intelligent Models



Understanding AI Cameras



Теоретическая основа

Машинное обучение в компьютерном зрении объединяет математику, статистику, оптимизацию и искусственный интеллект, чтобы компьютеры могли распознавать закономерности на изображениях.

В отличие от традиционного программирования:

Традиционное программирование

Входные данные + правила → выходные данные

Машинное обучение

Входные данные + желаемый результат → Машинное обучение → Модель

Модель изучает закономерности, а не программируется вручную.

Основные математические принципы:

- Линейная алгебра
- Математический анализ
- Теория вероятностей
- Статистика
- Оптимизация
- Матричные операции
- Нейронные сети

Данные изображения состоят из пикселей, расположенных в виде матриц.

Пример:

Изображение в оттенках серого:

1		28 × 28 pixels
2		
3		784 numerical values

Цветное изображение:

1		224 × 224 × 3
2		
3		Red
4		

Каждый пиксель становится числовым входным параметром для модели машинного обучения.

Определение

Практическое применение машинного обучения в компьютерном зрении — это процесс разработки интеллектуальных систем, которые автоматически анализируют, классифицируют, обнаруживают, сегментируют или интерпретируют визуальную информацию с помощью алгоритмов машинного обучения.

Его цели включают в себя:

- Классификация изображений
- Обнаружение объектов
- Сегментация изображений
- Распознавание лиц
- Оценка позы
- OCR (оптическое распознавание символов)
- Медицинская диагностика
- Промышленный инжиниринг
- Видеоаналитика

Пошаговый рабочий процесс разработки

Шаг **1** Определение проблемы

Прежде чем собирать данные, точно определите, чего должна достичь модель.

Примеры:

- Обнаружение поврежденных продуктов
- Подсчет транспортных средств
- Классификация растений
- Обнаружение опухолей
- Считывание номерных знаков

Without a clear objective, model performance becomes difficult to evaluate.

Step 2 Collect Image Data

Data sources may include:

- Company cameras
- Smartphones
- Industrial cameras
- Medical scanners
- Satellites
- Public datasets

High-quality datasets should contain:

✓ Different lighting

🔄 Multiple backgrounds

✓ Various camera angles

✓ Different object sizes

Step 3 Label the Dataset

Labels tell the algorithm the correct answer.

Examples:

Image	Label
Dog	Animal
Cat	Animal
Car	Vehicle
Apple	Fruit

For object detection:

Bounding boxes identify object locations.

For segmentation:

Every pixel receives its own class.

Step 4 Preprocess Images

Typical preprocessing includes:

- Resize images
- Normalize pixel values
- Remove noise
- Data augmentation
- Color correction
- Image cropping

Data augmentation improves generalization.

Examples:

- Rotation
- Flipping
- Zoom
- Brightness adjustment
- Random crop
- Blur

Step 5 Choose a Machine Learning Model

Popular models include:

Traditional ML

- SVM
- Random Forest
- KNN

Deep Learning

- CNN
- ResNet
- EfficientNet
- MobileNet
- Vision Transformer (ViT)
- YOLO
- Mask R-CNN

CNN remains the foundation of modern computer vision.

Step 6 Train the Model

Training involves:

Forward Pass

↓

Loss Calculation

↓

Backpropagation

↓

Weight Update

↓

Repeat

Thousands of iterations improve accuracy.

Step 7 Evaluate Performance

Common evaluation metrics include:

Classification

- Accuracy
- Precision
- Recall
- F1 Score

Detection

- mAP
- IoU

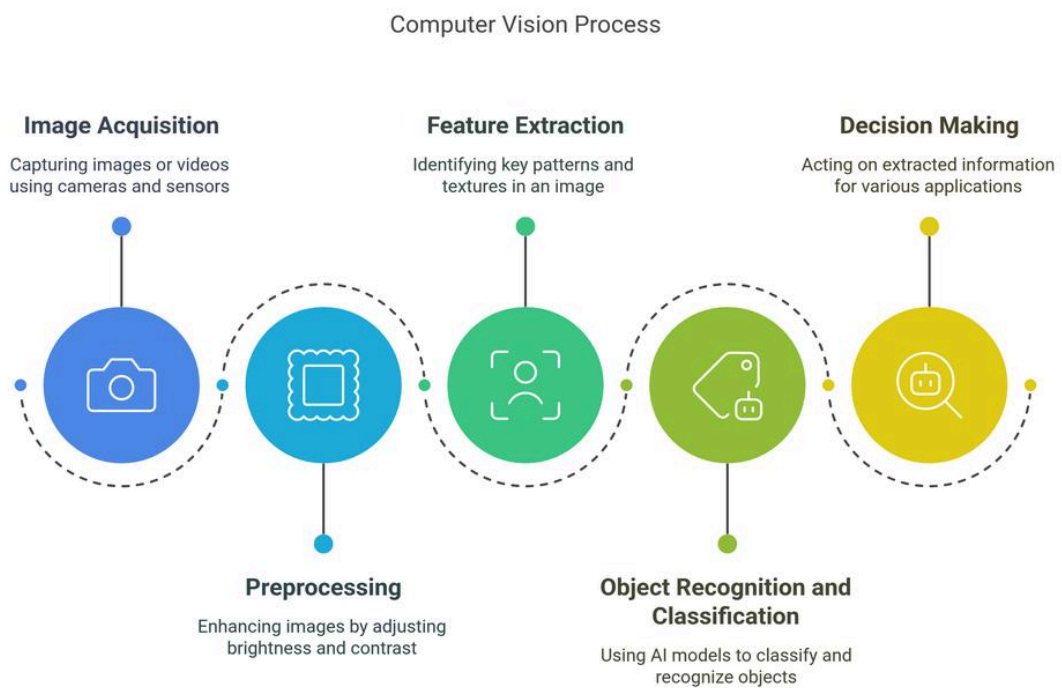
Segmentation

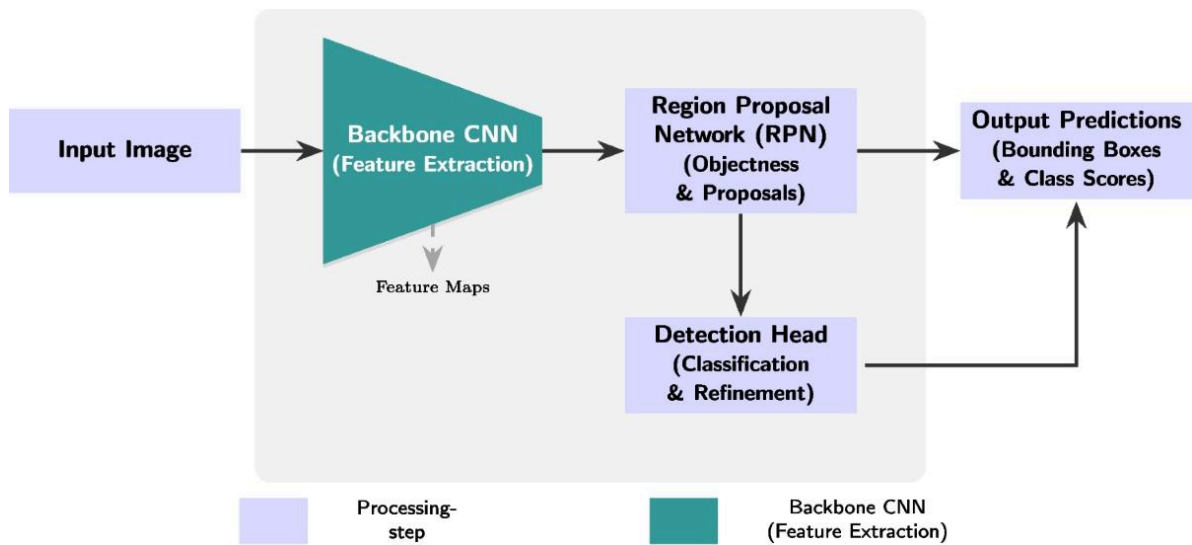
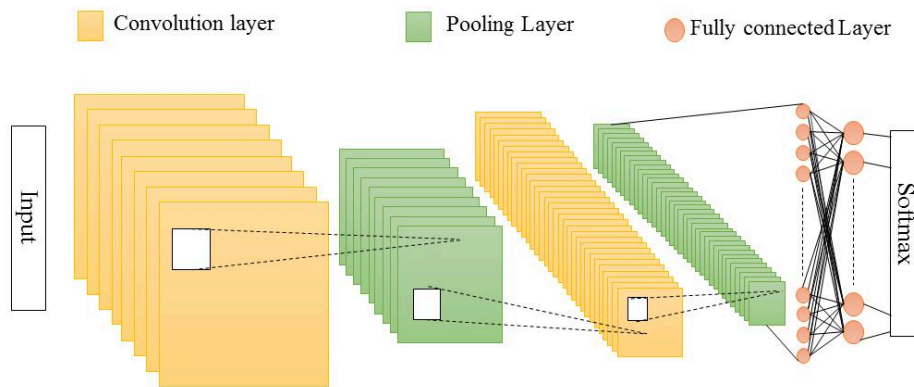
- Dice Score
- Pixel Accuracy
- MeanIoU

Step 8 Deploy the Model 🌐

Deployment options:

- Cloud
- Mobile Apps
- Raspberry Pi
- NVIDIA Jetson
- Industrial PLC
- Edge AI Devices



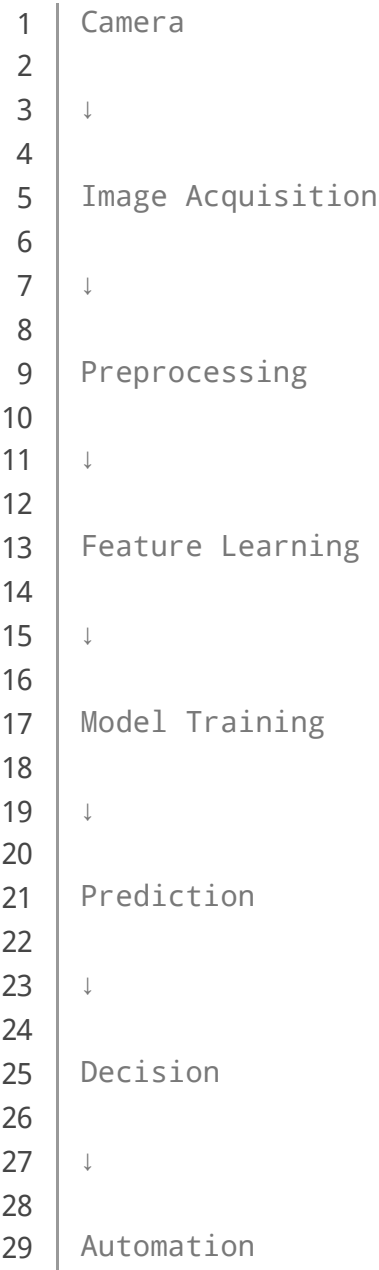


Comparison

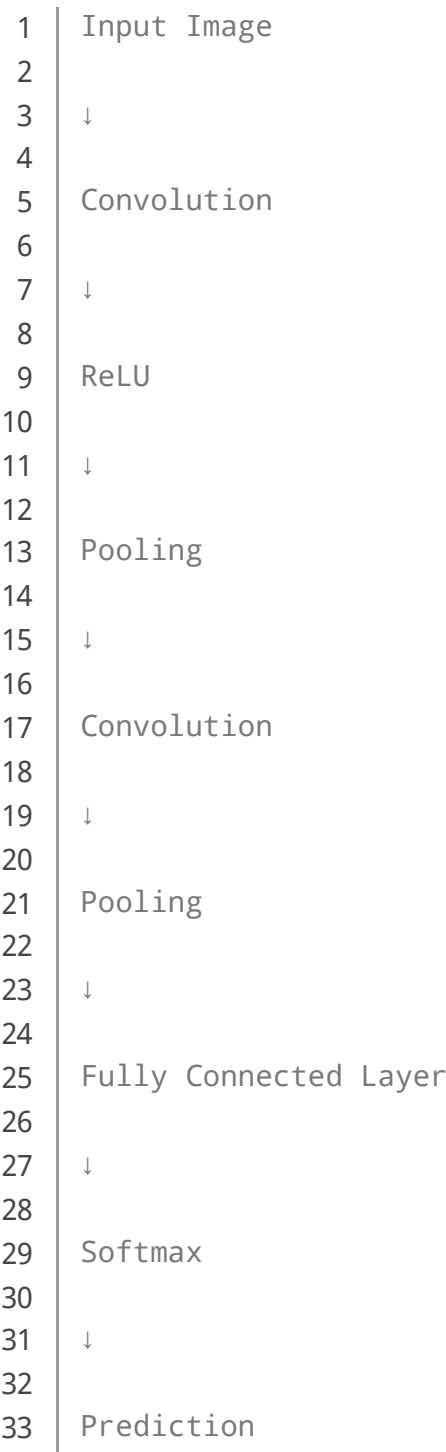
Feature	Traditional Image Processing	Machine Learning	Deep Learning
Rule Based	✓	✗	✗
Learns Automatically	✗	✓	✓
Large Dataset Required	✗	Medium	High
Accuracy	Medium	High	Very High

Feature	Traditional Image Processing	Machine Learning	Deep Learning
Feature Engineering	Manual	Partial	Automatic
Hardware Requirement	Low	Medium	High
Scalability	Medium	High	Very High

Computer Vision Pipeline Diagram



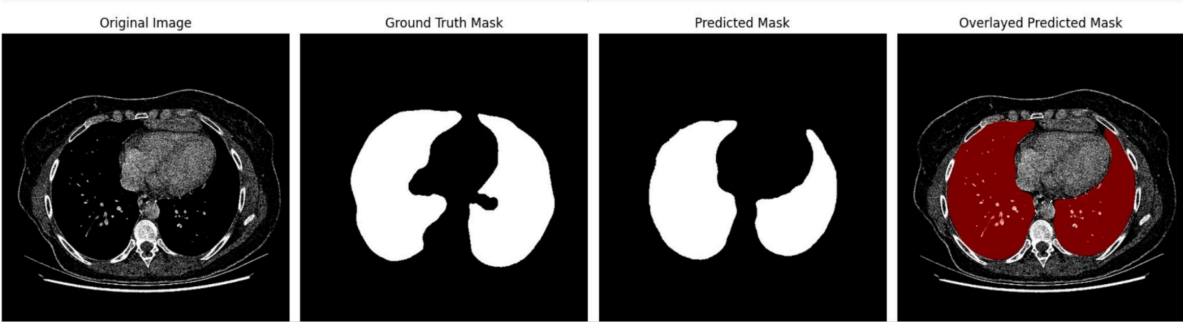
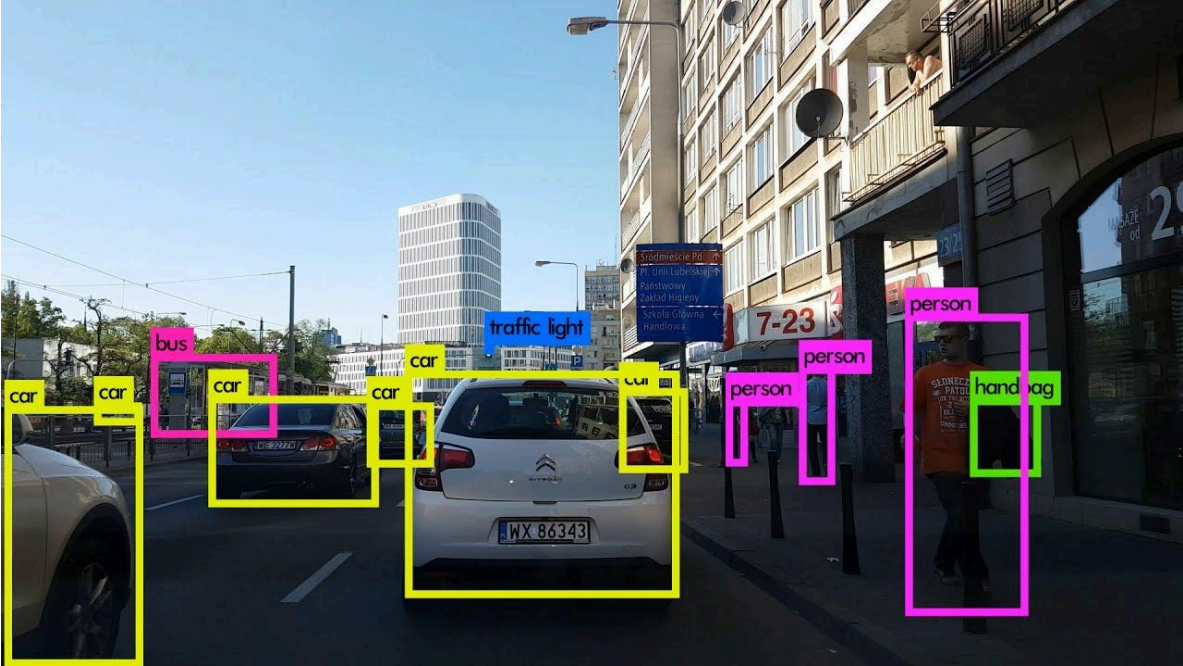
CNN Architecture Diagram



Popular Vision Models Comparison

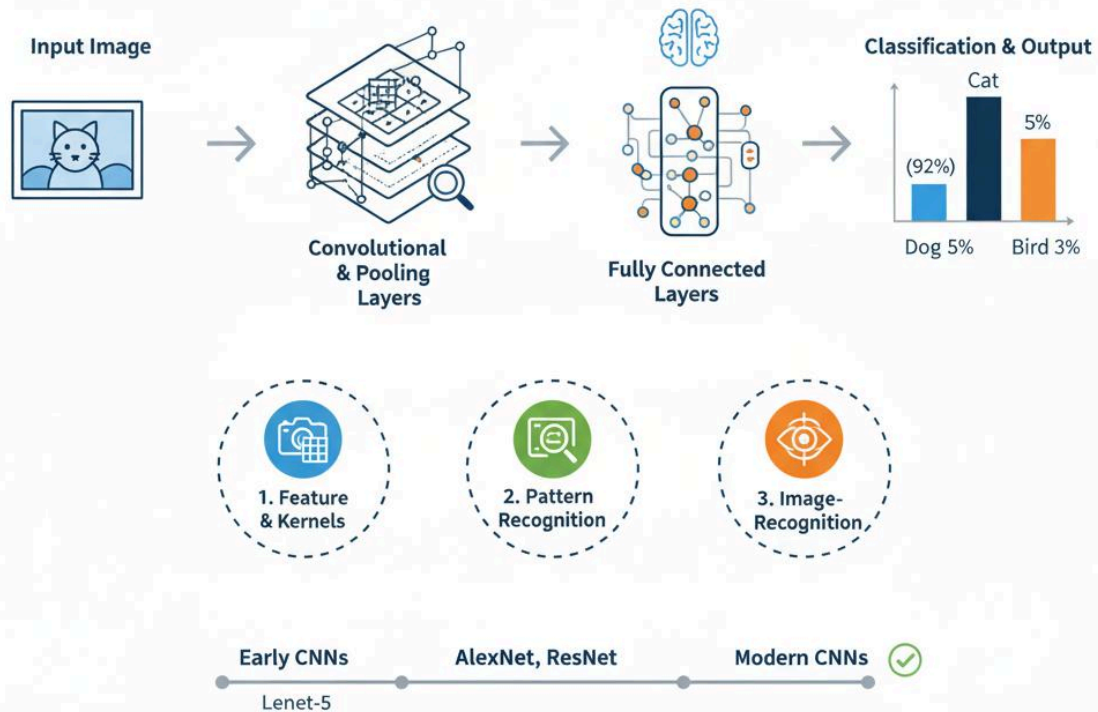
Model	Speed	Accuracy	Best For
LeNet	Fast	Low	Education
AlexNet	Medium	Medium	Learning CNN

Model	Speed	Accuracy	Best For
VGG16	Slow	High	Research
ResNet50	Medium	Excellent	General Vision
MobileNet	Very Fast	High	Mobile Devices
EfficientNet	High	Excellent	Production
YOLOv8	Extremely Fast	Excellent	Real-time Detection
Vision Transformer	Medium	Excellent	Large Datasets



Convolutional Neural Networks (CNNs): The Vision Pioneers

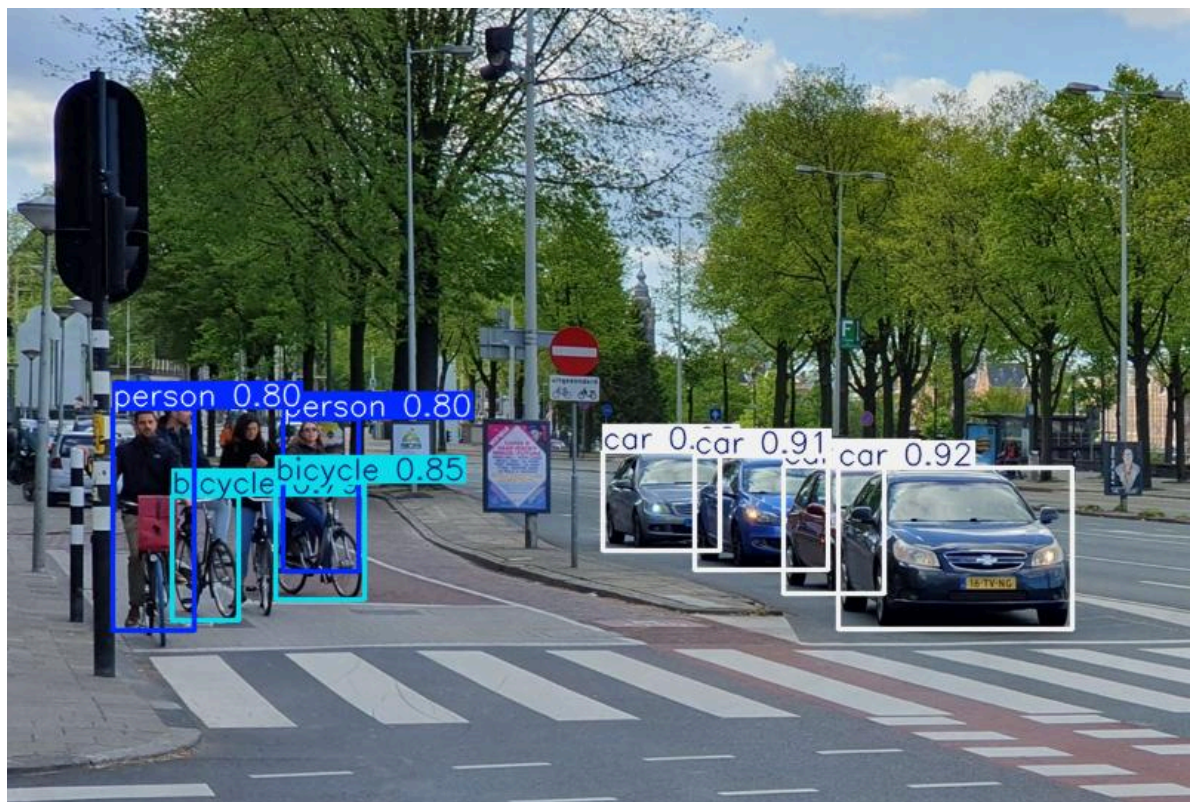
Unlocking Image Recognition & Computer Vision with Deep Learning

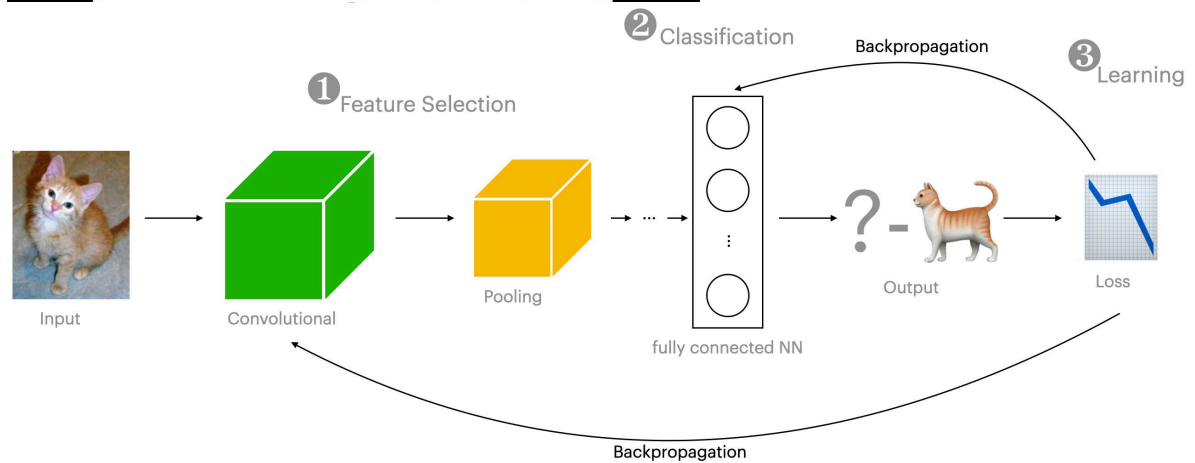
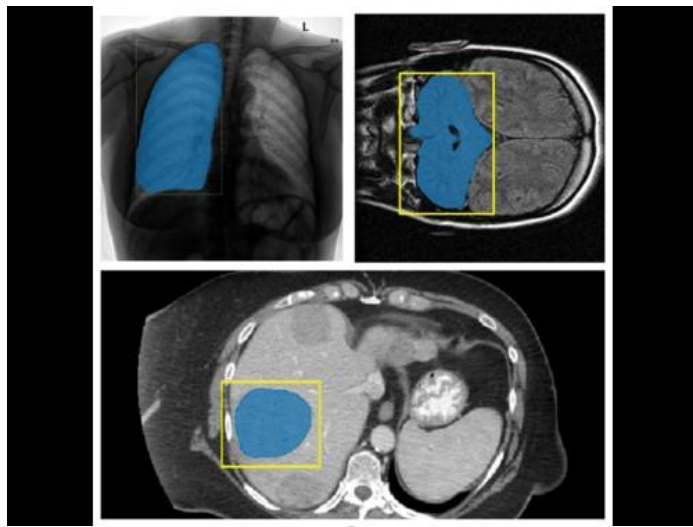


✓ Revolutionized Computer Vision

✓ Image Classification & Object Detection

✓ Medical Imaging & Autonomous Driving





Examples

Image Classification

Input:

Apple image

Output:

Apple

Confidence:

99.3%

Object Detection

Image:

Street

Detected Objects:

- Car
- Bicycle
- Pedestrian
- Traffic Light

Face Recognition

Applications:

- Smartphone Unlock
- Airport Security
- Employee Attendance

Medical Imaging

Detect:

- Lung Cancer
- Brain Tumors
- Fractures
- Skin Diseases

OCR

Convert:

Scanned Documents

↓

Editable Text

Real-World Applications

Machine learning powers computer vision across numerous industries.

Healthcare

- Disease detection
- MRI analysis
- CT scan interpretation
- Surgical robotics

Manufacturing

- Quality inspection
- Defect detection
- Assembly verification
- Predictive maintenance

Automotive

- Self-driving vehicles
- Lane detection
- Traffic sign recognition
- Pedestrian detection

Agriculture

- Crop monitoring
- Weed detection
- Fruit counting
- Precision farming

Retail

- Inventory management
- Customer analytics
- Automated checkout
- Shelf monitoring

Security

- Video surveillance
- Intrusion detection
- Face recognition
- Access control

Robotics

- Autonomous navigation

- Object grasping
- Warehouse automation
- Human-robot collaboration

Environmental Monitoring

- Wildlife tracking
- Forest monitoring
- Disaster assessment
- Ocean pollution detection

Common Mistakes

Many engineering teams encounter similar pitfalls when developing computer vision systems.

- Using too little training data.
- Ignoring class imbalance.
- Training with poor-quality or mislabeled images.
- Skipping data augmentation.
- Overfitting by training for too many epochs.
- Choosing overly complex models for edge devices.
- Evaluating models only with accuracy instead of precision, recall, or mAP.
- Deploying models without testing under real lighting and environmental conditions.

Avoiding these mistakes improves reliability, reduces maintenance costs, and accelerates deployment.

Challenges & Solutions

Challenge	Solution
Limited Data	Data augmentation and transfer learning
Overfitting	Regularization and dropout
Slow Training	GPU acceleration
Class Imbalance	Weighted loss functions

Challenge	Solution
Poor Lighting	Image enhancement
Real-Time Constraints	Lightweight models such as MobileNet or YOLO
Hardware Limits	Model pruning and quantization
Labeling Costs	Semi-supervised and active learning

Case Study

Automated Defect Detection in Manufacturing

A factory producing metal components experienced inconsistent manual inspections, resulting in defective parts reaching customers.

Project Goal

Develop a computer vision system capable of detecting surface defects in real time.

Solution

- Installed high-resolution industrial cameras.
- Collected thousands of images under different lighting conditions.
- Labeled defects such as scratches, dents, and cracks.
- Applied data augmentation to improve model robustness.
- Fine-tuned a pretrained convolutional neural network using transfer learning.
- Deployed the optimized model on an edge AI device connected to the production line.

Results

Metric	Before AI	After AI
Inspection Accuracy	88%	98.5%
Inspection Time	12 seconds	Less than 1 second
Production Defects	High	Reduced by over 60%
Labor Cost	High	Lower due to automation

This case demonstrates how practical machine learning can improve quality, reduce costs, and increase production efficiency.

Essential Tips

-  Start with well-curated datasets before experimenting with advanced models.
-  Use transfer learning to reduce training time and improve accuracy.
-  Keep separate training, validation, and test datasets to avoid data leakage.
-  Monitor loss curves and evaluation metrics throughout training.
-  Optimize models with pruning or quantization for edge deployment.
-  Select hardware that matches your performance and latency requirements.
-  Protect datasets containing sensitive information through anonymization and secure storage.
-  Continuously monitor deployed models and retrain them when data distributions change.

Frequently Asked Questions

What is Computer Vision?

Computer Vision is a branch of artificial intelligence that enables computers to interpret and analyze images and videos, allowing automated understanding of visual information.

Is Python the best language for Computer Vision?

Yes. Python is widely used because of its extensive ecosystem, including libraries such as OpenCV, TensorFlow, PyTorch, and scikit-learn.

What is the difference between [Machine Learning](#) and Deep Learning?

Machine Learning often requires manual feature engineering, while Deep Learning automatically learns hierarchical features directly from image data using neural networks.

Why are Convolutional Neural Networks (CNNs) so popular?

CNNs efficiently capture spatial patterns, edges, textures, and shapes, making them highly effective for image classification, object detection, and segmentation tasks.

Can Computer Vision work in real time?

Yes. With optimized models such as YOLO or MobileNet running on GPUs or edge AI hardware, many applications achieve real-time inference suitable for robotics, surveillance, and autonomous systems.

What hardware is recommended for training?

Dedicated GPUs significantly reduce training time. For large-scale projects, multiple GPUs or cloud-based accelerators are commonly used.

How much data is needed?

The required dataset size depends on task complexity. Transfer learning can achieve strong results with a few thousand labeled images, while training models from scratch often requires hundreds of thousands or millions of images.

Conclusion

Practical Machine Learning for Computer Vision has become a cornerstone of modern engineering, enabling machines to interpret visual data with speed and precision that support industries ranging from healthcare and manufacturing to transportation and agriculture. By following a structured workflow—defining the problem, collecting and labeling high-quality data, preprocessing images, selecting suitable models, training effectively, evaluating with meaningful metrics, and deploying optimized solutions—engineers can build reliable and scalable vision systems.

As hardware accelerators, edge AI devices, and advanced architectures continue to evolve, computer vision applications will become faster, more energy efficient, and increasingly accessible. Engineers and students who master both the theoretical foundations and practical implementation techniques will be well positioned to develop innovative AI solutions that improve safety, productivity, and decision-making across the USA, UK, Canada, Australia, Europe, and beyond.

Related Posts:



[Preview PDF Book](#)

[← Previous Book](#)

[Next Book →](#)

EngiVerse

Explore a vast library of free engineering ebooks curated to help you expand your technical knowledge, conduct groundbreaking research, and advance your career. Our mission is to create a hub of knowledge that fosters innovation, empowers aspiring engineers, and bridges the gap between education and industry.

[Home](#) [Books](#) [About Us](#) [Contact](#)
[Privacy Policy](#) [DMCA](#)

